Supplementary Materials

How does familiarity impact the computational processes that support face recognition?

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A1 Appendix A: Supplementary Materials

A1.1 Parameter recovery study

We ran a parameter recovery study that included 30 participant datasets, each with 180 trials (90 trials "familiar" and 90 trials "unfamiliar") and with our selected model parameterisation. Code can found in the parameter_recovery.qmd file in the /exp_1/ subfolder. For more details on parameter recovery studies, see Heathcote et al. (2019). The hierarchical model showed good recovery of the parameter value (top left panel), but other aspects of the model, such as the precision of the estimates, were less well-calibrated Figure A1.1. The example in the figure below is for one parameter, but all model parameters showed the same general pattern.

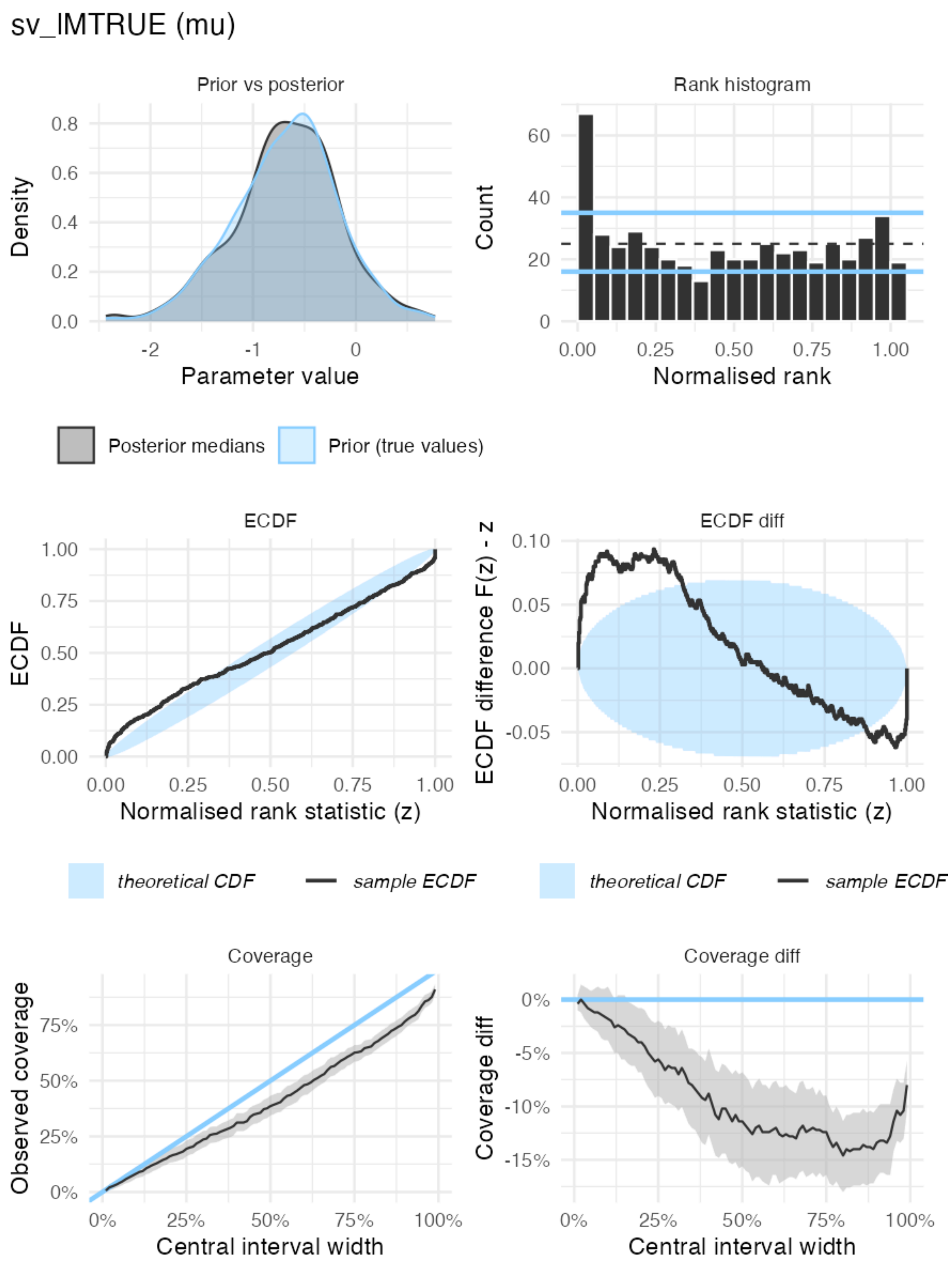


Figure A1.1. Parameter recovery for the hierarchical design in Experiment 1

We then ran a parameter recovery study for a simpler, non-hierarchical model, where each participant is modelled separately. The simpler model showed better overall recovery Figure A1.2. Importantly, when we fit the single model to our data, we got the same pattern of results as the hierarchical model. This gives us confidence that we can have reasonable trust in our model estimates, whilst also recognising that collecting more data would sharpen the precision of our estimates when fitting hierarchical models.

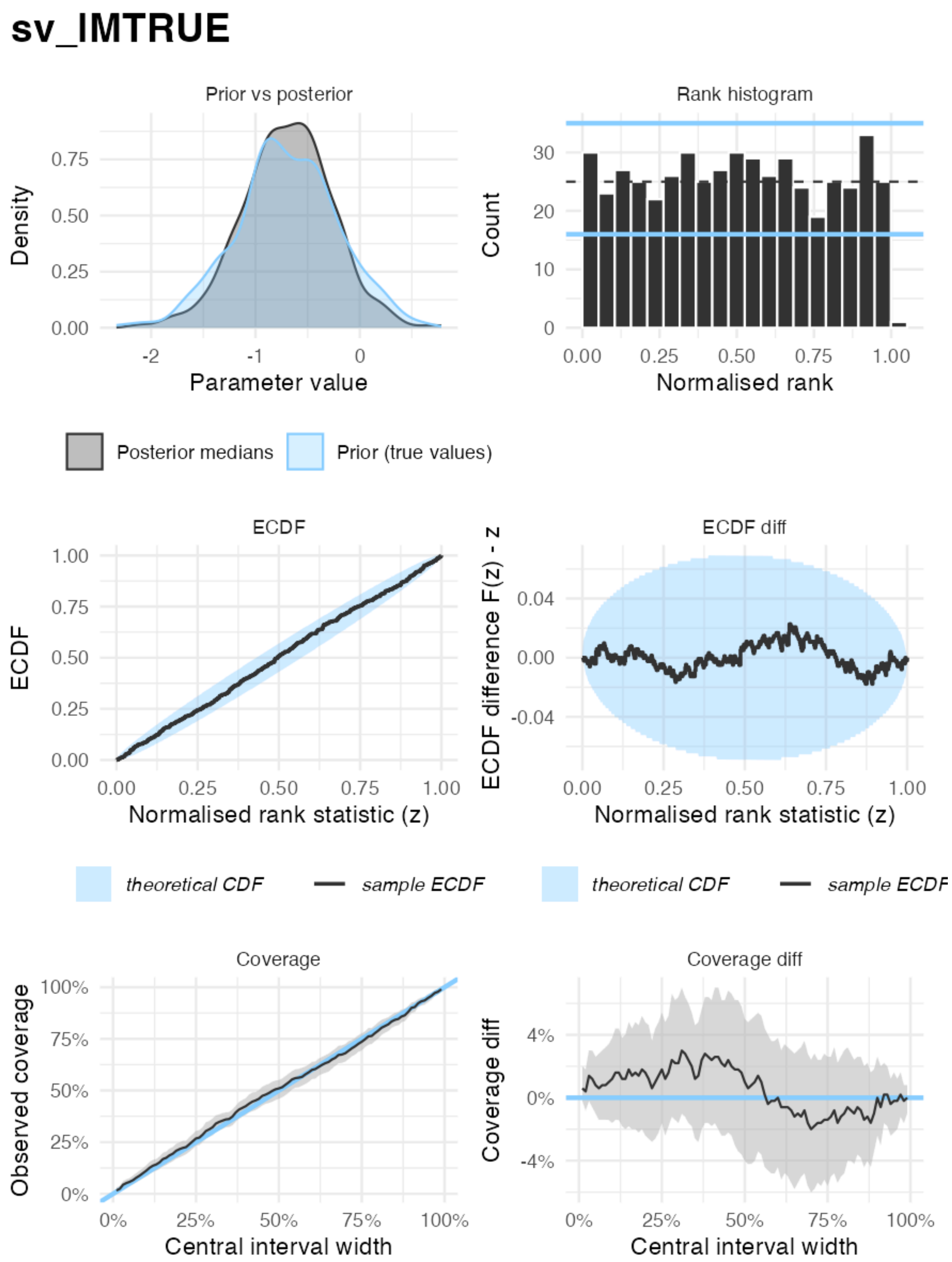


Figure A1.2. Parameter recovery for the single level design in Experiment 1

A1.2 Priors

These are the priors used for our pre-registered models. The example parameter names here are for Experiments 1 and 2 (i.e., Novel vs Learned), but these priors were used in all experiments.

parameter	description	condition	correct	response	mean	sd
v_SNovel:lMFALSE	Drift rate	Novel	False	-	0	2

parameter	description	condition	correct	response	mean	sd
v_SLearned:lMFALSE	Drift rate	Learned	False	-	0	2
v_SNovel:lMTRUE	Drift rate	Novel	True	-	1	2
v_SLearned:lMTRUE	Drift rate	Learned	True	-	1	2
sv_lMTRUE	SD drift rate (true accumulator)	-	True	-	log(0.5)	0.5
B_lRNovel	Threshold	-	-	Novel	log(1)	0.5
B_lRLearned	Threshold	-	-	Learned	log(1)	0.5
A	Start point	-	-	-	log(0.5)	0.3
t0	Non-decision time	-	-	-	log(0.3)	0.3

A1.3 Model checks

A1.3.1 Convergence

The chains showed satisfactory mixing as evidenced by the rhat values.

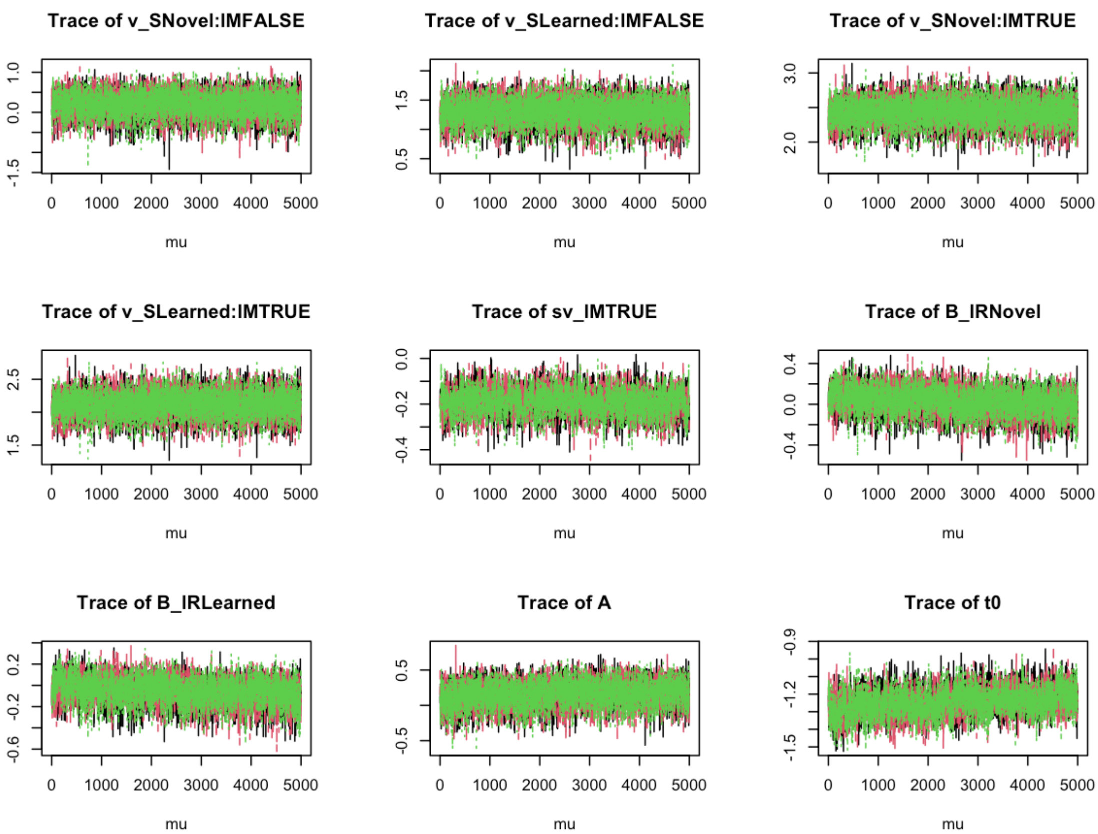
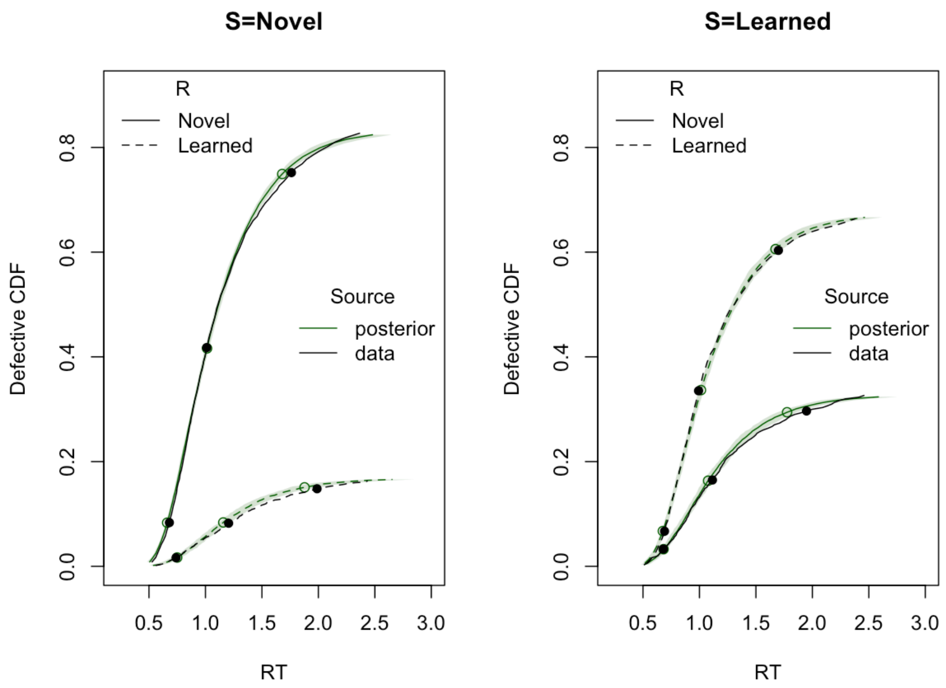
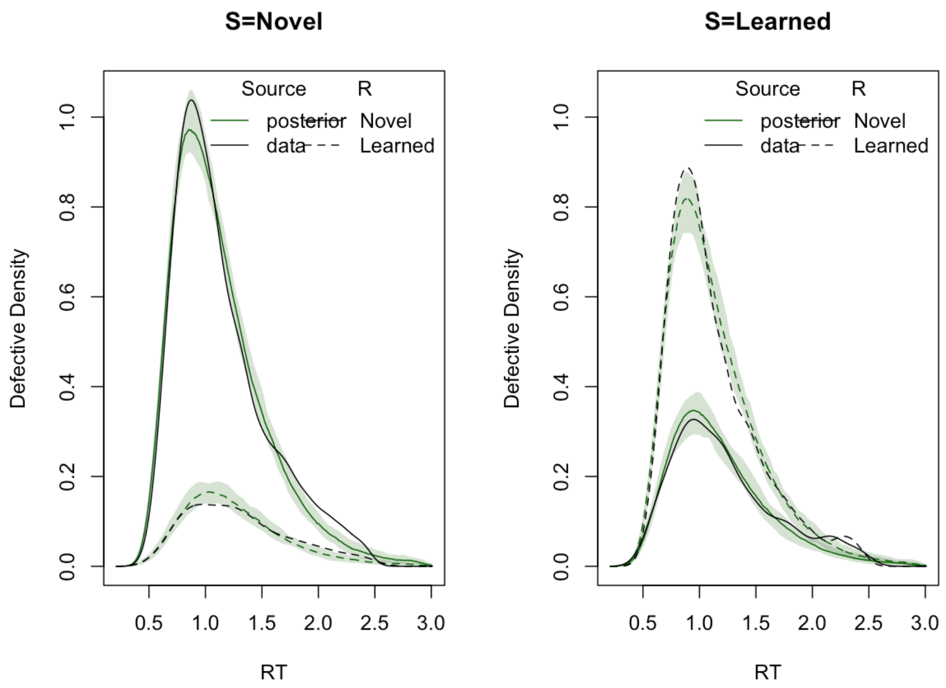


Figure A1.3. MCMC chains for Experiment 1

A1.3.2 Posterior predictive checks



(a) CDF



(b) Density

Figure A1.4. Posterior predictive checks for Experiment 1

A1.3.3 Prior vs posterior plots

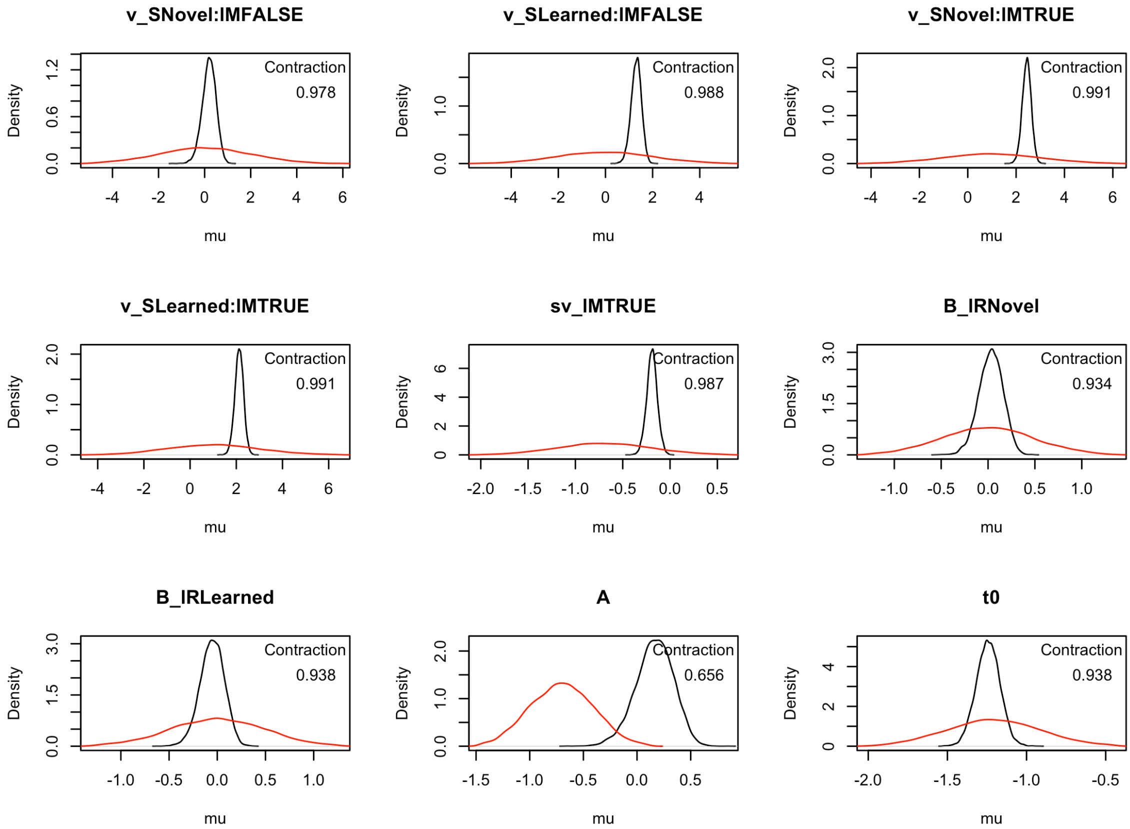


Figure A1.5. Priors vs posterior plots for Experiment 1

A1.4 Exploratory models

For each experiments where an LBA model was fitted (Experiment 1, 2, 4 and 5), exploratory models were fitted as well. Three exploratory models were fitted for each experiment, where we allowed other parameters from the LBA model to vary in addition to the two preregistered parameters (Drift rate and Threshold). Start point (A) was allowed to vary based on the response (Learned vs Novel for Experiment 1 and 2, Famous vs Novel for Experiment 4 and 5) and non-decision time (t0) were allowed to vary based on the stimulus condition (Learned vs Novel for Experiment 1 and 2, Famous vs Novel for Experiment 4 and 5).

Table A1.1. Formula specification for the exploratory LBA models.

(a) Start point models

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	yes	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	no	no	no	-

(b) Non-decision time models

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	no	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	yes	no	no	-

(c) Full models

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	yes	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	yes	yes	no	-
Non-decision time	t0	yes	no	no	-

References

Heathcote, A., Lin, Y.-S., Reynolds, A., Strickland, L., Gretton, M., & Matzke, D. (2019). Dynamic models of choice. *Behavior Research Methods*, *51*(2), 961–985. https://doi.org/10.3758/s13428-018-1067-y